

Absolutely. Based on your requirements and the architecture described in your existing agreement (AI inference engine + web interface), I would actually recommend evolving it into an **enterprise-grade AI Platform Functional Specification** rather than a simple requirements document.

The document should be written specifically so AI coding agents (Claude Code, GPT-5, Codex, Cursor, Windsurf, Cline, etc.) can build each module independently while maintaining a clean architecture.

I would structure it as a comprehensive Software Requirements Specification (SRS) in Markdown of approximately **120–180 pages** containing:

1. Executive Summary

- Project Overview
 - Objectives
 - Business Goals
 - Scope
 - Success Criteria
 - Assumptions
 - Constraints
-

2. System Architecture

Including architecture diagrams for:

- AI Training Pipeline
 - AI Inference Platform
 - Multi-tenant SaaS Architecture
 - API Gateway
 - Authentication
 - Model Registry
 - Storage
 - GPU Workers
 - Report Engine
 - Pricing Engine
 - Audit Logs
-

3. Technology Stack

Example:

Frontend

- Next.js
- Tailwind
- React
- TypeScript

Backend

- FastAPI
- Python
- PostgreSQL
- Redis
- Celery

AI

- PyTorch
- YOLOv11
- SAM2
- MONAI
- EfficientNet
- Detectron2

Infrastructure

- Docker
- Kubernetes
- NVIDIA Triton
- MLflow
- DVC

Cloud

- AWS / Azure

4. AI Pipeline

Detailed specifications for

Dataset

Annotation

Training

Validation

Inference

Deployment

Model Monitoring

Model Drift

Continuous Retraining

5. AI Detection Modules

Each detection module described separately.

Example

Tooth Detection

Inputs

Outputs

Confidence

Training Dataset

Metrics

Acceptance Criteria

Inference Pipeline

Repeat for

- Caries
- Crown
- Missing Tooth
- Root Canal
- Implant Detection
- Sinus Detection
- Inferior Alveolar Nerve
- Periapical Lesions
- Bone Loss
- Bone Density

- Prosthesis
- Wisdom Teeth
- Orthodontic Appliances

6. Clinical Decision Engine

Rules engine

Example

IF

Missing Tooth

Bone Height > Threshold

No Sinus Lift

Recommend Implant

Estimated Cost

USD xxxx

Confidence

92%

7. Treatment Recommendation Engine

Maps

Detection

↓

Clinical Rule

↓

Treatment

↓

Estimated Duration



Estimated Cost



Priority



Risk

8. Quotation Engine

Automatically generate

Treatment

Quantity

Unit Price

Total

Optional Procedures

Mandatory Procedures

VAT

Currency

Insurance

Discount

PDF

9. Multi-Tenant SaaS

Organization

Clinic

Users

Roles

Permissions

Branding

Pricing

Billing

Tenant Isolation

API Keys

Custom Domains

10. Authentication

Admin

Clinic Owner

Dentist

Operator

Sales Agent

Super Admin

JWT

OAuth

MFA

11. Case Management

Every uploaded case receives

Case ID

Clinic ID

Patient ID

Upload Date

Status

Review

Version

Audit Trail

12. Workflow

Upload X-ray



AI Queue



Inference



Treatment Engine



Quotation



Dentist Review



Approval



Patient Quote

13. AI REST APIs

Including

Upload API

Inference API

Cases API

Users API

Reports API

Pricing API

Tenant API

Authentication API

Webhooks

14. Database Design

Complete ERD

Including

Users

Tenants

Cases

Images

Treatments

Models

Inference Jobs

Audit Logs

Pricing Tables

Reports

15. AI Coding Agent Development Rules

This section is extremely important.

Example

Every coding agent must

- Write typed code
- 100% test coverage
- SOLID principles
- Repository pattern
- Hexagonal architecture
- Dependency Injection
- Clean Architecture
- Feature folders
- No business logic in controllers
- API First
- OpenAPI generation
- Complete documentation

16. Repository Structure

```
apps/  
  web  
  api  
  ai-training  
  ai-inference  
  
packages/  
  auth  
  ui  
  common  
  pricing  
  clinical  
  ai  
  reporting  
  
services/  
  inference  
  quotation  
  notifications  
  
infra/  
  docker  
  kubernetes  
  terraform
```



17. Prompt Library for AI Coding Agents

Example prompts

Build Authentication Module

Build AI Inference Module

Build Report Engine

Build Tenant Management

Build Pricing Engine

Build PDF Generator

Build DICOM Loader

Build GPU Worker

Build Monitoring Dashboard

18. Phased Development Plan

I would recommend **14 implementation phases** instead of trying to build everything at once.

Phase	Deliverable
Phase 1	Architecture, Repository, CI/CD
Phase 2	Authentication & Multi-Tenant Framework
Phase 3	Dataset Management & Annotation Platform
Phase 4	AI Training Pipeline
Phase 5	Tooth Detection Model
Phase 6	Disease Detection Models
Phase 7	Implant Analysis Models
Phase 8	Clinical Decision Engine

Phase	Deliverable
Phase 9	Pricing & Quotation Engine
Phase 10	AI Inference API
Phase 11	Next.js Web Portal
Phase 12	Reporting & PDF Generation
Phase 13	MLOps, Monitoring & Security
Phase 14	Production Deployment, Validation & Go-Live

Each phase would include:

- Functional requirements
- User stories
- Acceptance criteria
- Technical tasks
- AI coding agent prompts
- Testing requirements
- Deliverables
- Exit criteria

19. Testing Strategy

- Unit Testing
- Integration Testing
- Clinical Validation
- Model Validation
- Performance Testing
- Load Testing
- Security Testing
- Penetration Testing
- UAT
- Regression Testing

20. DevOps & MLOps

- GitHub Actions

- Docker
- Kubernetes
- Model Registry
- MLflow
- DVC
- Canary Deployment
- GPU Autoscaling
- Model Versioning
- Rollback Strategy
- Monitoring
- Drift Detection

Recommended Deliverables

Rather than a single markdown file, I recommend producing a complete development documentation set:

1. **Functional Specification** (`01_Functional_Specification.md`) — comprehensive business and functional requirements.
2. **Software Architecture Specification** (`02_Software_Architecture.md`) — system architecture, infrastructure, APIs, and data flow.
3. **AI Model Development Specification** (`03_AI_Model_Development.md`) — dataset requirements, training pipeline, evaluation metrics, and deployment.
4. **API Specification** (`04_API_Specification.md`) — OpenAPI-style endpoint definitions and payloads.
5. **Database Design Specification** (`05_Database_Design.md`) — schema, ERD, indexes, and tenant isolation.
6. **Development Roadmap** (`06_Implementation_Plan.md`) — phased implementation plan, milestones, sprint planning, and acceptance criteria.
7. **AI Coding Agent Playbook** (`07_AI_Coding_Agent_Guide.md`) — coding standards, reusable prompts, development workflow, testing expectations, and repository conventions.

This documentation set would provide a complete blueprint for an AI-assisted engineering team to implement the platform consistently, with clear separation of responsibilities between AI model development, backend services, frontend application, infrastructure, and MLOps. It would also scale well as additional dental detection modules or new clinical capabilities are introduced in future releases.